CSC 2114 Assignment 3: Weighted Campus Route Finder (UCS and IDS)

Group 10B · Deliverable: `10B_route_finder.py`

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1. Problem formulation (Part A)

The state space S is the set of the 18 places (14 schools, 4 landmarks); a state is the place where the visitor stands.

1. **Initial state** (an element of S): the source place, e.g. s_0 = School of Law. *Unchanged.*
2. **Actions(s)** (a function S → set of actions): {Go(n) : n is a neighbour of s}. The weighted adjacency structure `{place: {neighbour: weight}}` is built from `WEIGHTED_EDGES` in both directions; the actions are its keys, in alphabetical order. *Unchanged in form*; the weights play no part in which actions are legal.
3. **Transition model Result(s, a)** (a function S × A → S): Result(s, Go(n)) = n. Deterministic. *Unchanged.*
4. **Goal test** (a Boolean test on a state): IsGoal(s) ⇔ s = destination. *Unchanged.*
5. **Step and path cost** (functions to the positive integers): c(s, Go(n), n) = `graph[s][n]`, the walking distance of the edge in metres; the path cost is the sum of step costs. *Changed*: in Assignment 1 every step cost 1. Every step cost is positive because every weight is a positive whole number (the smallest is 70 m) and `build_graph` rejects any weight ≤ 0.

What the weights keep ignoring. (1) Slope: Makerere is on a hill, and walking up costs more effort than walking down. (2) Crowds at lecture change-over. (3) Rain and time of day. Slope can be folded in without changing the form: c(s, a, s′) = w(s, s′) × (1 + k·climb(s, s′)), which is still a positive function of the edge. It would make edges directed (A → B ≠ B → A), but every component keeps its type.

Why positive step costs. With a 0-weight edge, UCS can keep pushing nodes that walk back and forth along it without g growing, so in tree-like mode there would be infinitely many nodes with g below the goal's cost and UCS might never reach the goal; ties also stop meaning "equal distance". With a negative edge, a loop through it lowers g every time, so there is no cheapest route at all, and the guarantee that the first time UCS removes a node its g is minimal fails, because a later path could still get cheaper.

State versus node, revisited. Searching from School of Law, the state Main Library first appears in a node reached via Economics with g = 120 + 400 = 520 m, and then in a node reached via Freedom Square with g = 180 + 90 = 270 m (trace in Section 3). UCS keeps the 270 m node; the 520 m node becomes stale.

What IDS needs from a node. DEPTH, the number of hops from the root. A child gets it from its parent: `depth = parent.depth + 1` (0 for the root).

Which quantity is minimised. BFS minimises hops. UCS minimises metres. IDS also minimises hops (it returns the first goal at the smallest depth limit); none of BFS, DFS or IDS minimises metres.

2. Design: what changed since Assignment 1

- **Weighted graph.** `build_graph` returns `{place: {neighbour: weight}}` with neighbours in alphabetical order; `step_cost(graph, s, s2)` reads one weight. Every search function receives the graph as an argument.
- **Node** has STATE, PARENT, ACTION, PATH-COST (metres) and a fifth field DEPTH = parent.depth + 1. Hops reported are the depth of the goal node. `solution()` is unchanged; `route_cost(graph, route)` recomputes a route's cost from the graph, and the printed cost comes from it.
- **reached** now holds the *cheapest* node found so far for each state.
- **Frontiers.** `FIFOFrontier` (BFS), `LIFOFrontier` (DFS; children pushed in reverse so the alphabetically first is explored first) and `PriorityFrontier` (UCS, a `heapq` of `(path_cost, state_name, counter, node)` with `counter` from `itertools.count()`: cheapest first, then alphabetical, then earliest inserted; two nodes are never compared).
- **Reached rule** (a `reopens` flag on each strategy): BFS and DFS add a child only if its state is not in reached; UCS also adds it if it reaches the state with a strictly lower PATH-COST, replacing the stored node. **Stale rule:** when UCS removes a node whose g exceeds the g stored in reached, it is discarded silently and not counted as expanded.
- **IDS.** The driver `ids_search` calls the *same* generic `search` with limits 0, 1, 2, …, passing the stack (DFS) frontier, reached off, the cycle check on (a child whose place is already on the path from the root is not generated)

and `depth_limit = L`. A node at depth L is goal-tested but not expanded and not counted, and sets a *cutoff* flag; an iteration with no cutoff means the whole space was searched, so the driver prints *No route found.* The limit never exceeds 17 (places − 1). Expansions are summed over all iterations.

- **Interface.** The prompt accepts BFS, DFS, UCS, IDS and ALL (case-insensitive, trimmed). Unknown places or algorithms print a message and the valid choices; source equal to destination prints a one-place route with cost 0, hops 0 and 0 nodes expanded. Flags: `--no-reached`, `--trace`, `--verbose`.

```
search(graph, source, goal, strategy, use_reached):
    frontier ← strategy.make_frontier()                    # plug-in 1: FIFO | LIFO | heap ordered by g
    frontier.add([root]);  reached ← {source: root}        # reached ← None in tree-like mode
    while frontier not empty:
        node ← frontier.pop()
        if strategy.reopens and node.g > reached[node.state].g:  continue   # stale (UCS)
        if node.state = goal:  return node                  # goal test on REMOVAL
        if node.depth = depth_limit:  cutoff ← true; continue    # plug-in 3 (IDS only)
        if expanded = 10,000:  return CAP
        expanded ← expanded + 1
        for child in expand(graph, node):                   # alphabetical
            if cycle_check and child.state on path(node):  drop  # IDS only
            seen ← reached[child.state]
            if seen and not (strategy.reopens and child.g < seen.g):  drop   # plug-in 2: reached rule
            else: reached[child.state] ← child; keep child
        frontier.add(kept children)
    return FAILURE
ids(graph, source, goal):  for L = 0 … 17:  r ← search(graph, source, goal, DFS, use_reached=False,
                                                       depth_limit=L, cycle_check=True)
                           stop when r found the goal, or when r.cutoff is false (No route found.)
```

3. Verification (B4)

Pair	UCS required (m / hops / exp.)	UCS obtained	IDS required	IDS obtained	Result
EDU → ECO	550 / 4 / 8	550 / 4 / 8	560 / 2 / 4	560 / 2 / 4	Pass
CoCIS → LPA	820 / 4 / 13	820 / 4 / 13	840 / 3 / 11	840 / 3 / 11	Pass
SBE → LAW	870 / 5 / 11	870 / 5 / 11	1170 / 5 / 32	1170 / 5 / 32	Pass

Watching reached being updated (UCS, School of Law → School of the Built Environment; full trace in Appendix B):

```
pop School of Economics key=120
push Main Library g=520 key=520
pop Freedom Square key=180
replace Main Library g=520 -> 270
…
pop Main Library key=520
stale Main Library
```

Main Library is first reached from Economics along the 400 m edge, so it enters reached and the heap with g = 520. Freedom Square (g = 180) is removed next and offers Main Library over a 90 m edge, g = 270. Because 270 < 520, the reached entry is replaced and a new node is pushed. The 520 m node was not good enough because it was reached through Economics, which is cheap to reach (120 m) but connected to the Library by a long edge; UCS found that first only because it expands in order of g, not because it was the best way in. When the old node is finally removed, its g (520) exceeds the stored 270, so it is discarded as stale and not counted. The same happens to Margaret Trowell (1100 → 920 m).

Tree-like UCS. School of Education → School of Economics: 35 expansions without reached (against 8 with reached), same 550 m route. School of Statistics and Planning → School of Veterinary Medicine: 10000 expansions, *Search stopped: expansion cap (10000) reached.* (against 17 with reached). Without reached, every walk is a separate node, so UCS must expand every walk with g below the goal's cost, including all walks that go round cycles such as the Freedom Square–Main Library–Main Building triangle (90, 120, 150 m). For a 550 m goal there are few such walks; for an 1840 m goal their number grows exponentially and passes 10,000 first. Without the cap UCS would still stop: every step costs at least 70 m, so only finitely many walks have g below 1840 m, and once they are exhausted the goal node is removed.

IDS without reached. Each iteration is finite: the depth limit bounds the length of every path, and the cycle check forbids repeating a place on the current path, so no path can loop. The limit grows only while something was cut off, and the longest loop-free path has 17 hops, so IDS terminates. What it gives up is memory of other branches: the same place is re-expanded via different paths in the same iteration (125 expansions at limit 8 on an 18-place graph).

4. Results (Part C)

Ten pairs, all four algorithms, graph search (reached on), ALL mode. Each cell is **cost (m) / hops / nodes expanded**; the full 40-row table with routes is in Appendix A. Every cost was checked against `route_cost`.

#	Source	Destination	BFS	DFS	UCS	IDS
1	ECO	LAW	120 / 1 / 2	120 / 1 / 16	120 / 1 / 1	120 / 1 / 1
2	MT	ENG	140 / 1 / 1	140 / 1 / 1	140 / 1 / 2	140 / 1 / 1
3	SBE	LAW	1170 / 5 / 12	1170 / 5 / 9	870 / 5 / 11	1170 / 5 / 32
4	EDU	ECO	560 / 2 / 7	560 / 2 / 16	550 / 4 / 8	560 / 2 / 4
5	CoCIS	LPA	840 / 3 / 11	820 / 4 / 4	820 / 4 / 13	840 / 3 / 11
6	SS	ENG	1140 / 4 / 14	920 / 5 / 5	920 / 5 / 13	1140 / 4 / 24
7	CONAS	SBE	1480 / 6 / 17	1240 / 7 / 9	1240 / 7 / 16	1480 / 6 / 61
8	SSP	VET	2050 / 7 / 17	2120 / 9 / 13	1840 / 9 / 17	2050 / 7 / 275
9	LAW	VET	1570 / 7 / 17	1780 / 8 / 15	1570 / 7 / 17	1570 / 7 / 221
10	MT	VET	2250 / 8 / 17	2340 / 11 / 11	2130 / 10 / 17	2250 / 8 / 324

Condition	Count	Pairs (rows of the 40-row table)
UCS returns a cheaper route than BFS	7 pairs	pair 3 (rows 9, 11); pair 4 (rows 13, 15); pair 5 (rows 17, 19); pair 6 (rows 21, 23); pair 7 (rows 25, 27); pair 8 (rows 29, 31); pair 10 (rows 37, 39)
UCS's route has more hops than BFS's	6 pairs	pair 4 (rows 13, 15); pair 5 (rows 17, 19); pair 6 (rows 21, 23); pair 7 (rows 25, 27); pair 8 (rows 29, 31); pair 10 (rows 37, 39)
IDS costs more metres than UCS	7 pairs	pair 3 (rows 11, 12); pair 4 (rows 15, 16); pair 5 (rows 19, 20); pair 6 (rows 23, 24); pair 7 (rows 27, 28); pair 8 (rows 31, 32); pair 10 (rows 39, 40)
UCS expands more nodes than BFS	3 pairs	pair 2 (rows 5, 7); pair 4 (rows 13, 15); pair 5 (rows 17, 19)
UCS expands fewer nodes than BFS	4 pairs	pair 1 (rows 1, 3); pair 3 (rows 9, 11); pair 6 (rows 21, 23); pair 7 (rows 25, 27)
IDS expands ≥ 5× as many nodes as BFS	3 pairs	pair 8 (rows 29, 32); pair 9 (rows 33, 36); pair 10 (rows 37, 40)
Directly connected schools	2 pairs	pair 1 (rows 1); pair 2 (rows 5)
Fewest-hops distance ≥ 6	4 pairs	pair 7 (rows 25); pair 8 (rows 29); pair 9 (rows 33); pair 10 (rows 37)

--verbose on the deepest pair (pair 10, MT → VET, 8 hops by BFS): Max frontier size BFS 5, DFS 6, UCS 4, IDS 12. IDS per iteration: Depth limit 0: 0 expanded, cutoff · Depth limit 1: 1 expanded, cutoff · Depth limit 2: 4 expanded, cutoff · Depth limit 3: 8 expanded, cutoff · Depth limit 4: 14 expanded, cutoff · Depth limit 5: 27 expanded, cutoff · Depth limit 6: 51 expanded, cutoff · Depth limit 7: 94 expanded, cutoff · Depth limit 8: 125 expanded, goal found

One terminal run of ALL (the program also offers GBFS and A* from Assignment 4; those two blocks are omitted here):

```
$ python3 route_finder.py -s "School of Education" -d "School of Economics" -a ALL

Algorithm: BFS
Route: School of Education -> Main Library -> School of Economics
Cost (metres): 560
Hops: 2
Nodes expanded: 7

Algorithm: DFS
Route: School of Education -> Main Library -> School of Economics
Cost (metres): 560
Hops: 2
Nodes expanded: 16

Algorithm: UCS
Route: School of Education -> Main Library -> Freedom Square -> School of Law -> School of Economics
Cost (metres): 550
Hops: 4
Nodes expanded: 8

Algorithm: IDS
Route: School of Education -> Main Library -> School of Economics
Cost (metres): 560
Hops: 2
Nodes expanded: 4
Final depth limit: 2
```

5. Analysis

1. UCS optimality. UCS returns a cheapest route when (a) every step cost is at least some ε > 0 (here ε = 70 m) and (b) the goal test happens when a node is *removed* from the frontier, not when it is generated. Pair 4 (EDU → ECO): EDU is expanded, then SS (140), then ML (160), which generates ECO along the 400 m edge at g = 560. A UCS that goal-tested at generation would stop there and return EDU → ML → ECO, 560 m (the BFS route). Our UCS only queues ECO at 560, keeps expanding cheaper nodes (FS 250, LAW 430), finds ECO again via Law at 550, replaces it, and removes the 550 m node first: EDU → ML → FS → LAW → ECO, 550 m.

2. BFS stops being optimal. Pair 3 (SBE → LAW, figure). BFS returns SBE → MT → LPA → MB → FS → LAW = 70 + 450 + 350 + 120 + 180 = **1170 m**; UCS returns SBE → ENG → CoCIS → CTF → FS → LAW = 90 + 250 + 130 + 220 + 180 = **870 m**. Both have 5 hops. BFS measures only depth, so at depth 5 the two routes are indistinguishable to it; it returns whichever it reaches first, and the alphabetical rule makes it go through Margaret Trowell ("M" before "School of Engineering"), right into the two longest edges on campus (450 m and 350 m). It cannot see the 300 m difference because it never reads a weight. Pairs 4–8 and 10 show the same failure.

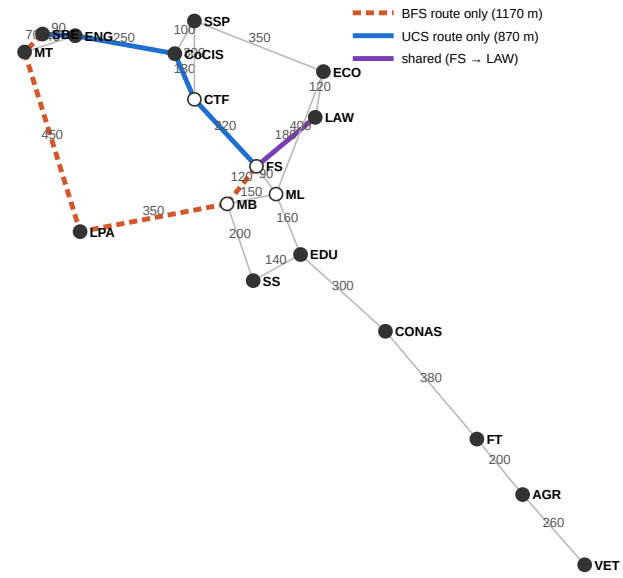


Figure 1. Pair 3, SBE → LAW, on the map coordinates, with edge weights in metres.

3. IDS versus BFS and UCS. IDS is never cheaper than UCS: UCS is optimal in metres. IDS returned exactly the BFS route in all ten pairs, and its cost equals UCS's only when the fewest-hops route also happens to be the cheapest (pairs 1, 2 and 9, e.g. LAW → VET, 1570 m). Whenever the cheapest route needs more hops (pairs 4–8, 10) or a different route of equal hops (pair 3), IDS costs more: by up to 300 m (pair 3) and 240 m (pair 7).

4. Where IDS's repeated work goes. For pair 10, IDS expanded 0, 1, 4, 8, 14, 27, 51, 94 and 125 nodes at limits 0–8: **324** in total, against 17 for BFS (19×). The lecture's argument is that in a tree with branching factor b the last level holds most of the nodes, so repeating the shallower levels costs only about a factor b/(b−1) extra. That holds when b is large. Here the average degree is only 48/18 ≈ 2.7 (about 1.7 new choices after excluding the parent), so the shallow levels are a large share of the work. Worse, the graph is heavily cyclic and IDS has no reached table: within one iteration it re-expands the same place along many different loop-free paths, so the final iteration alone does 125 expansions on an 18-place graph. BFS expands each place at most once.

5. Memory. For pair 10: UCS 4 < BFS 5 < DFS 6 < IDS 12. All are small because the graph has 18 places and low degree. BFS and UCS frontiers grow at the hubs (MB, FS, ML with four neighbours each), which put several new places on the frontier at once, but reached stops any place entering twice (UCS only re-adds a place when it is strictly cheaper). DFS keeps the siblings of every node on its current path. IDS is largest here because it has no reached table, so duplicates of the same place along different paths sit on the stack together. The lecture's point is asymptotic: IDS needs only O(b·d) memory, against $O(b^d)$ for BFS; on a large tree IDS would win clearly. Note too that BFS and UCS also hold the reached table (up to 18 entries), which IDS does not.

6. Two ways to avoid loops. The reached table remembers every place seen, so it needs memory for all reached states, but it prevents any place from being explored twice across the whole search, and its replace rule is what lets UCS re-open a state at a lower cost. In BFS and DFS it can wrongly throw away a later, cheaper path to a state (keeping the first one), which is harmless for hops but is exactly the bug UCS's replace rule fixes. The cycle check needs only the current path (O(d) memory) and can only throw away paths that revisit a place, which are never optimal with positive costs, but it does not stop the same place from being reached by different branches, which is where IDS's repeated work comes from.

7. Do the uninformed algorithms read the weights? *Prediction*: dropping MT–LPA from 450 m to 150 m cannot change any BFS, DFS or IDS route, because they never read weights; it changes the reported cost of any of their

routes that use that edge. UCS can change its route, because the edge is now cheaper. *Result* over all 306 ordered pairs:

Algorithm	Route changed	Same route, new cost	Unaffected
BFS	0	51	255
DFS	0	42	264
UCS	42	10	254
IDS	0	51	255

The prediction holds. Example, pair 6 (SS → ENG): BFS keeps its route through LPA and MT, whose cost falls from 1140 to 840 m; UCS abandons its 920 m route and switches to SS → MB → LPA → MT → ENG (840 m).

8. Limitation of UCS. For pair 9 (LAW → VET), UCS made 17 expansions, and 8 of them were places whose cheapest distance to VET is *larger* than from LAW itself (1570 m): ECO, CTF, SSP, CoCIS, LPA, ENG, SBE, MT. That is the whole west side of campus, in the opposite direction from the Veterinary school. UCS expands in rings of equal g around the source and has no idea where the goal is. The best-first algorithm that avoids this is **A*** ($f = g + h$); it needs a heuristic h(n), an estimate of the remaining distance to the goal, for example the straight-line distance computed from map coordinates of each place.

6. Reflection

We would make the weights depend on direction: on Makerere hill, walking up from the Agricultural and Veterinary schools takes far longer than walking down. That changes only the step and path cost component ($c(s, a, s') \neq c(s', a', s)$), and the graph would have to store each direction separately. The most surprising result for us was how badly IDS's repeated work behaved: the lecture's "small overhead" became 19 times BFS's work on our deepest pair, because the campus is small, low-branching and full of cycles.

Appendix A – Results table (40 rows)

MB Main Building, FS Freedom Square, ML Main Library, CTF Central Teaching Facility, ENG Engineering, SBE Built Environment, MT Margaret Trowell, LAW Law, EDU Education, SS Social Sciences, LPA Liberal and Performing Arts, ECO Economics, SSP Statistics and Planning, CONAS Natural Sciences, FT Food Technology, AGR Agricultural Sciences, VET Veterinary Medicine.

#	Source	Destination	Alg.	Route	Cost (m)	Hops	Nodes exp.
1	School of Economics	School of Law	BFS	ECO → LAW	120	1	2
2	School of Economics	School of Law	DFS	ECO → LAW	120	1	16
3	School of Economics	School of Law	UCS	ECO → LAW	120	1	1
4	School of Economics	School of Law	IDS	ECO → LAW	120	1	1
5	Margaret Trowell School of Industrial and Fine Art	School of Engineering	BFS	MT → ENG	140	1	1
6	Margaret Trowell School of Industrial and Fine Art	School of Engineering	DFS	MT → ENG	140	1	1
7	Margaret Trowell School of Industrial and Fine Art	School of Engineering	UCS	MT → ENG	140	1	2
8	Margaret Trowell School of Industrial and Fine Art	School of Engineering	IDS	MT → ENG	140	1	1
9	School of the Built Environment	School of Law	BFS	SBE → MT → LPA → MB → FS → LAW	1170	5	12
10	School of the Built Environment	School of Law	DFS	SBE → MT → LPA → MB → FS → LAW	1170	5	9
11	School of the Built Environment	School of Law	UCS	SBE → ENG → CoCIS → CTF → FS → LAW	870	5	11
12	School of the Built Environment	School of Law	IDS	SBE → MT → LPA → MB → FS → LAW	1170	5	32
13	School of Education	School of Economics	BFS	EDU → ML → ECO	560	2	7
14	School of Education	School of Economics	DFS	EDU → ML → ECO	560	2	16
15	School of Education	School of Economics	UCS	EDU → ML → FS → LAW → ECO	550	4	8
16	School of Education	School of Economics	IDS	EDU → ML → ECO	560	2	4
17	CoCIS	School of Liberal and Performing Arts	BFS	CoCIS → ENG → MT → LPA	840	3	11
18	CoCIS	School of Liberal and Performing Arts	DFS	CoCIS → CTF → FS → MB → LPA	820	4	4
19	CoCIS	School of Liberal and Performing Arts	UCS	CoCIS → CTF → FS → MB → LPA	820	4	13
20	CoCIS	School of Liberal and Performing Arts	IDS	CoCIS → ENG → MT → LPA	840	3	11
21	School of Social Sciences	School of Engineering	BFS	SS → MB → LPA → MT → ENG	1140	4	14
22	School of Social Sciences	School of Engineering	DFS	SS → MB → FS → CTF → CoCIS → ENG	920	5	5
23	School of Social Sciences	School of Engineering	UCS	SS → MB → FS → CTF → CoCIS → ENG	920	5	13
24	School of Social Sciences	School of Engineering	IDS	SS → MB → LPA → MT → ENG	1140	4	24
25	College of Natural Sciences	School of the Built Environment	BFS	CONAS → EDU → ML → MB → LPA → MT → SBE	1480	6	17
26	College of Natural Sciences	School of the Built Environment	DFS	CONAS → EDU → ML → FS → CTF → CoCIS → ENG → SBE	1240	7	9
27	College of Natural Sciences	School of the Built Environment	UCS	CONAS → EDU → ML → FS → CTF → CoCIS → ENG → SBE	1240	7	16
28	College of Natural Sciences	School of the Built Environment	IDS	CONAS → EDU → ML → MB → LPA → MT → SBE	1480	6	61
29	School of Statistics and Planning	School of Veterinary Medicine	BFS	SSP → ECO → ML → EDU → CONAS → FT → AGR → VET	2050	7	17
30	School of Statistics and Planning	School of Veterinary Medicine	DFS	SSP → CTF → FS → MB → SS → EDU → CONAS → FT → AGR → VET	2120	9	13
31	School of Statistics and Planning	School of Veterinary Medicine	UCS	SSP → CoCIS → CTF → FS → ML → EDU → CONAS → FT → AGR → VET	1840	9	17
32	School of Statistics and Planning	School of Veterinary Medicine	IDS	SSP → ECO → ML → EDU → CONAS → FT → AGR → VET	2050	7	275
33	School of Law	School of Veterinary Medicine	BFS	LAW → FS → ML → EDU → CONAS → FT → AGR → VET	1570	7	17
34	School of Law	School of Veterinary Medicine	DFS	LAW → FS → MB → SS → EDU → CONAS → FT → AGR → VET	1780	8	15

35	School of Law	School of Veterinary Medicine	UCS	LAW → FS → ML → EDU → CONAS → FT → AGR → VET	1570	7	17
36	School of Law	School of Veterinary Medicine	IDS	LAW → FS → ML → EDU → CONAS → FT → AGR → VET	1570	7	221
37	Margaret Trowell School of Industrial and Fine Art	School of Veterinary Medicine	BFS	MT → LPA → MB → ML → EDU → CONAS → FT → AGR → VET	2250	8	17
38	Margaret Trowell School of Industrial and Fine Art	School of Veterinary Medicine	DFS	MT → ENG → CoCIS → CTF → FS → MB → SS → EDU → CONAS → FT → AGR → VET	2340	11	11
39	Margaret Trowell School of Industrial and Fine Art	School of Veterinary Medicine	UCS	MT → ENG → CoCIS → CTF → FS → ML → EDU → CONAS → FT → AGR → VET	2130	10	17
40	Margaret Trowell School of Industrial and Fine Art	School of Veterinary Medicine	IDS	MT → LPA → MB → ML → EDU → CONAS → FT → AGR → VET	2250	8	324

Appendix B – UCS --trace, School of Law → School of the Built Environment

```
push School of Law g=0 key=0
pop School of Law key=0
push Freedom Square g=180 key=180
push School of Economics g=120 key=120
pop School of Economics key=120
push Main Library g=520 key=520
push School of Statistics and Planning g=470 key=470
pop Freedom Square key=180
replace Main Library g=520 -> 270
push Central Teaching Facility g=400 key=400
push Main Building g=300 key=300
push Main Library g=270 key=270
pop Main Library key=270
push School of Education g=430 key=430
pop Main Building key=300
push School of Liberal and Performing Arts g=650 key=650
push School of Social Sciences g=500 key=500
pop Central Teaching Facility key=400
push CoCIS g=530 key=530
pop School of Education key=430
push College of Natural Sciences g=730 key=730
pop School of Statistics and Planning key=470
pop School of Social Sciences key=500
pop Main Library key=520
stale Main Library
pop CoCIS key=530
push School of Engineering g=780 key=780
pop School of Liberal and Performing Arts key=650
push Margaret Trowell School of Industrial and Fine Art g=1100 key=1100
pop College of Natural Sciences key=730
push School of Food Technology, Nutrition and Bioengineering g=1110 key=1110
pop School of Engineering key=780
replace Margaret Trowell School of Industrial and Fine Art g=1100 -> 920
push Margaret Trowell School of Industrial and Fine Art g=920 key=920
push School of the Built Environment g=870 key=870
pop School of the Built Environment key=870
```